

HW9 - Computer Simulation II

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1

Using the histogram reweighting method we get

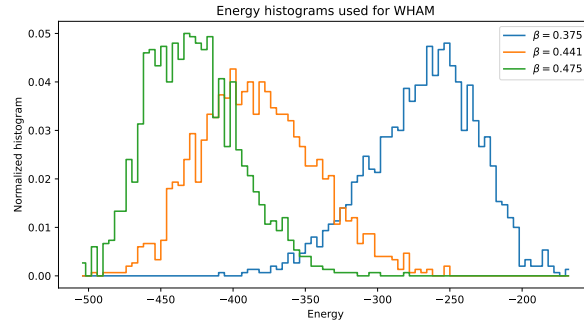


Figure 1: Energy histograms.

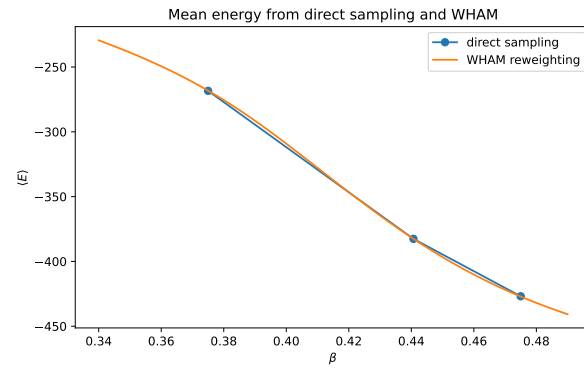


Figure 2: Mean energy from direct sampling and from WHAM.

The WHAM estimate agrees well with the direct measurements near the simulated values and provides a smooth interpolation of the energy curve between the three simulation points.

2

Using the improve estimator for the susceptibility we get:

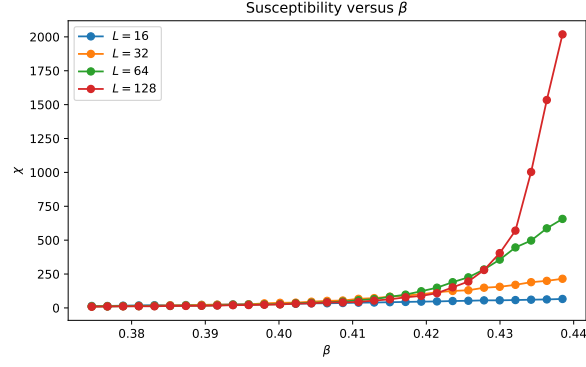


Figure 3: Susceptibility as a function of β for several lattice sizes.

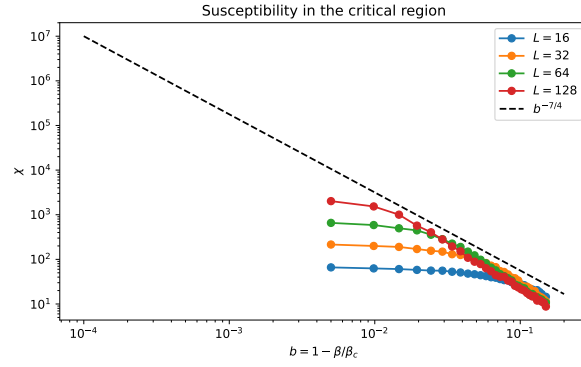


Figure 4: Log-log plot of χ versus $b = 1 - \beta/\beta_c$ with the expected power law $b^{-7/4}$.

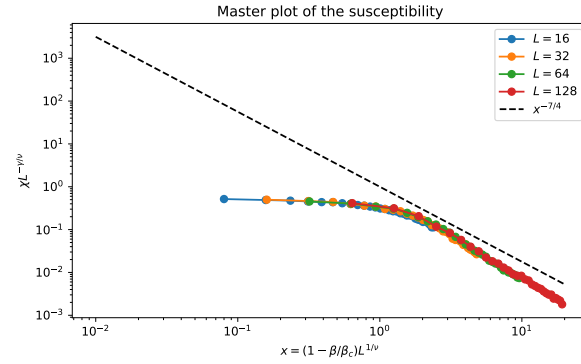


Figure 5: Master plot of $\chi L^{-\gamma/\nu}$ versus $x = (1 - \beta/\beta_c)L^{1/\nu}$, using $\nu = 1$ and $\gamma = 7/4$.

The susceptibility grows strongly near the critical point $\beta = 0.44$ for $L > 16$, scaling with lattice size. In the second plot we can see that the susceptibility approaches the expected near β_c but still shows big dependence on lattice size, while the master plot shows a lattice size independent result.